PCBC H Parametric & Sequences Test (Practice B) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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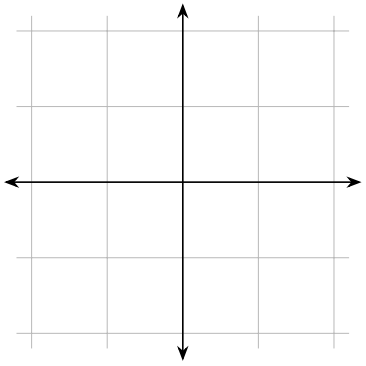
_____ _____

Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

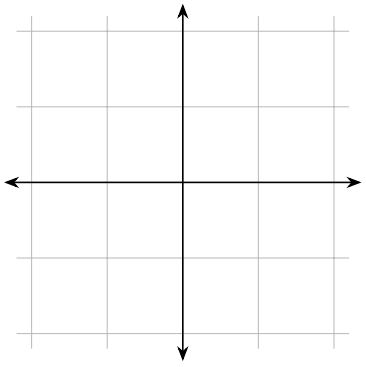
1. $x = 4 - 2t, \quad y = \dfrac{t^2}{2} + t, \quad -2 \le t \le 4$

SIMPLIFIED rectangular form: _____



2. $x = \sqrt{t-1}, \quad y = t + 2, \quad 1 \le t \le 5$

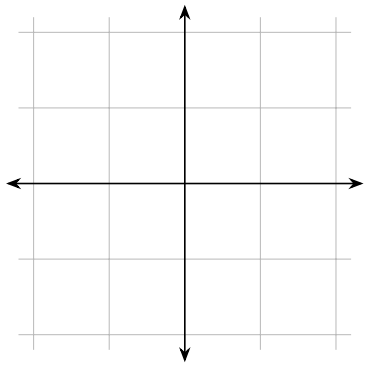
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

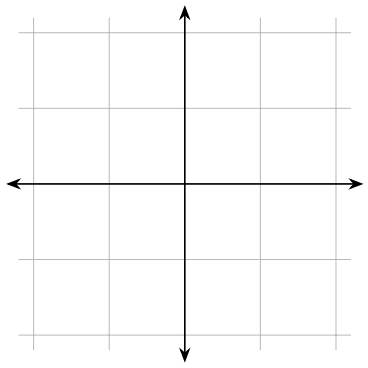
3. $x = 5\sin t, \quad y = -5\cos t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 4\tan t, \quad y = 3\sec t, \quad \dfrac{\pi}{2} < t < \dfrac{3\pi}{2}$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = 3(x+1)^2 - 6$

Write down the first five terms of the sequence. (2 points)

6. $\{a_n\} = \left\{(-1)^n \dfrac{n^2-1}{2n}\right\}$

Extra practice: the other type #6 could be

Not part of the 16 points. Worked in the key.

A sequence is defined recursively. Write the first five terms.

E1. $a_1 = 4; \quad a_n = \dfrac{a_{n-1}}{2n}$

Answer Key

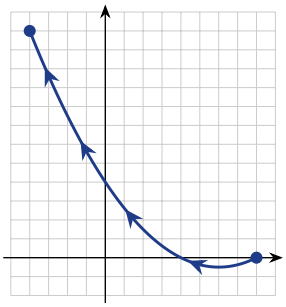
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 4 - 2t,\ y = \frac{t^2}{2} + t,\ -2 \le t \le 4$

t	(x,y)
-2	$(8,0)$
-1	$(6,-\frac{1}{2})$
0	$(4,0)$
2	$(0,4)$
4	$(-4,12)$

brackets: both ends



starts at $(8,0)$, moves right to left

$t = \frac{4-x}{2}$ — *isolate t*

$y = \frac{1}{2}\left(\frac{4-x}{2}\right)^2 + \frac{4-x}{2}$ — *substitute*

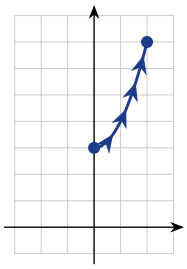
$y = \frac{16-8x+x^2}{8} + \frac{16-4x}{8}$ — *common denominator 8*

$\boxed{y = \frac{1}{8}x^2 - \frac{3}{2}x + 4 \ \text{ for } x \text{ in } [-4,8]}$ — *SIMPLIFIED: y alone*

Check: $x = 6$ gives $\frac{36}{8} - 9 + 4 = -\frac{1}{2}$. Matches $t = -1$. $y = \frac{x^2-12x+32}{8}$ is the same.

2. $x = \sqrt{t-1},\ y = t + 2,\ 1 \le t \le 5$

t	(x,y)
1	$(0,3)$
2	$(1,4)$
5	$(2,7)$



left to right

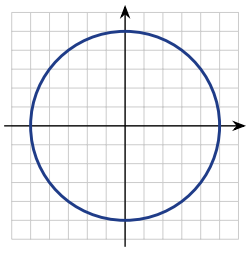
$x^2 = t - 1 \Rightarrow t = x^2 + 1$ — *square, then add 1*

$y = (x^2 + 1) + 2$ — *substitute*

$\boxed{y = x^2 + 3 \ \text{ for } x \text{ in } [0,2]}$ — *a square root is never negative*

3. $x = 5\sin t,\ y = -5\cos t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(0,-5)$
$\frac{\pi}{2}$	$(5,0)$
π	$(0,5)$
$\frac{3\pi}{2}$	$(-5,0)$
2π	$(0,-5)$



whole circle, radius 5

$\sin t = \frac{x}{5}, \quad \cos t = -\frac{y}{5}$

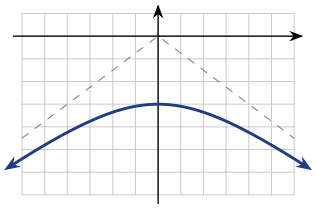
$\frac{x^2}{25} + \frac{y^2}{25} = 1$ — $\sin^2 t + \cos^2 t = 1$

$\boxed{x^2 + y^2 = 25}$ — *whole circle: no domain*

4. $x = 4\tan t,\ y = 3\sec t,\ \frac{\pi}{2} < t < \frac{3\pi}{2}$

t	(x,y)
$\frac{3\pi}{4}$	$(-4,-3\sqrt{2})$
π	$(0,-3)$
$\frac{5\pi}{4}$	$(4,-3\sqrt{2})$

$\sec t < 0$: *y negative*



bottom branch; end arrowheads

$\tan t = \frac{x}{4}, \quad \sec t = \frac{y}{3}$

$\frac{y^2}{9} - \frac{x^2}{16} = 1$ $\qquad \sec^2 t - \tan^2 t = 1$

$y^2 = 9\left(\frac{x^2}{16} + 1\right)$ $\qquad$ *bottom branch; a domain can't cut it: solve for y*

$\boxed{y = -3\sqrt{\frac{x^2}{16} + 1}}$ $\qquad$ *keep −; never "$y \le -3$"*

5. $y = 3(x+1)^2 - 6$

$\boxed{x = t,\ y = 3(t+1)^2 - 6}$

$x = t - 1:\ y = 3(t - 1 + 1)^2 - 6$ $\qquad$ *pick $t-1$ so the inside is t*

$\boxed{x = t - 1,\ y = 3t^2 - 6}$

Both for t in $(-\infty,\infty)$ (assumed; writing it is fine). Give both $x =$ and $y =$; don't expand.

6. $\left\{(-1)^n \frac{n^2-1}{2n}\right\}$

$(-1)^1\frac{0}{2},\ (-1)^2\frac{3}{4},\ (-1)^3\frac{8}{6},$ $\qquad$ *plug in $n = 1$ to 5*

$(-1)^4\frac{15}{8},\ (-1)^5\frac{24}{10}$

$\boxed{0,\ \frac{3}{4},\ -\frac{4}{3},\ \frac{15}{8},\ -\frac{12}{5}}$ $\qquad$ $\frac{0}{2} = 0$; *reduce* $\frac{8}{6}$ *and* $\frac{24}{10}$

Extra practice (not scored)

E1. $a_1 = 4;\ a_n = \frac{a_{n-1}}{2n}$

$a_2 = \frac{4}{4} = 1,\ \ a_3 = \frac{1}{6}$ $\qquad$ *divide by $2n$: 4, then 6, 8, 10*

$a_4 = \frac{1/6}{8} = \frac{1}{48},\ \ a_5 = \frac{1/48}{10} = \frac{1}{480}$

$\boxed{4,\ 1,\ \frac{1}{6},\ \frac{1}{48},\ \frac{1}{480}}$